Point-by-Point Response to Technical Check

Manuscript Title: Explainable Machine Learning-Based Diabetes Prediction Using Clinical Features: A Comparative Evaluation of Ensemble and Boosting Algorithms

Manuscript ID: b0d2381e-3d88-4943-9cdf-8be21a65c486 v1.0

Dear Editorial Office,

Thank you for the technical check of our manuscript. We have carefully addressed all requested points and uploaded a clean revised manuscript. Our responses are provided below.

1. Please provide City in the affiliation for author(s) (Aff1) in the manuscript.

Response: The city has been added to Affiliation 1. The affiliation now includes Dhaka.

2. Please provide Country affiliation for the author(s) (Aff1) in the manuscript.

Response: The country has been added to Affiliation 1. The affiliation now includes Bangladesh.

3. Please provide a Funding Declaration in the manuscript. If there was no Funding, please state so.

Response: A Funding Declaration has been added to the manuscript: "This research received no external funding."

4. The location of Table Citation (Table 1) is not clear in the manuscript file. Indicate in the manuscript where the table should appear.

Response: Table 1 is now explicitly cited in the Results section, and its intended placement is clearly indicated in the manuscript.

5. The location of Figure Citation (Fig. 1) is not clear in the manuscript. Ensure the figure is cited and indicate in the manuscript where it should appear.

Response: Figure 1 is now cited in the ROC-AUC Analysis section, and its intended placement is clearly indicated.

6. The location of Figure Citation (Fig. 2) is not clear in the manuscript. Ensure the figure is cited and indicate in the manuscript where it should appear.

Response: Figure 2 is now cited in the Confusion Matrix Analysis section, and its intended placement is clearly indicated.

7. The location of Figure Citation (Fig. 3) is not clear in the manuscript. Ensure the figure is cited and indicate in the manuscript where it should appear.

Response: Figure 3 is now cited in the Explainable AI-Based Feature Interpretation section, and its intended placement is clearly indicated.

8. The location of Figure Citation (Fig. 4) is not clear in the manuscript. Ensure the figure is cited and indicate in the manuscript where it should appear.

Response: Figure 4 is now cited in the Explainable AI-Based Feature Interpretation section, and its intended placement is clearly indicated.

9. The Citation for Reference (Cortes & Vapnik 1995) is missing in the manuscript.

Response: The in-text citation for Cortes and Vapnik (1995) has been added in the Introduction while discussing Support Vector Machines.

10. The Citation for Reference (International Diabetes Federation 2021) is missing in the manuscript.

Response: The in-text citation for the International Diabetes Federation (2021) has been added in the Introduction in the context of the global diabetes burden.

11. The Citation for Reference (Saeedi et al. 2019) is missing in the manuscript.

Response: The in-text citation for Saeedi et al. (2019) has been added in the Introduction in relation to global diabetes prevalence and projections.

12. The Citation for Reference (Hosmer et al. 2013) is missing in the manuscript.

Response: The in-text citation for Hosmer et al. (2013) has been added in the Logistic Regression subsection of the Methods.

13. The Citation for Reference (Molnar 2022) is missing in the manuscript.

Response: The in-text citation for Molnar (2022) has been added in the Introduction while discussing model interpretability and black-box machine learning.

14. The Citation for Reference (Sisodia & Sisodia 2018) is missing in the manuscript.

Response: The in-text citation for Sisodia and Sisodia (2018) has been added in the Introduction in the discussion of machine-learning approaches for diabetes prediction.

15. The Citation for Reference (Kavakiotis et al. 2017) is missing in the manuscript.

Response: The in-text citation for Kavakiotis et al. (2017) has been added in the Introduction in the discussion of machine learning in diabetes research.

16. The Citation for Reference (Zou et al. 2018) is missing in the manuscript.

Response: The in-text citation for Zou et al. (2018) has been added in the Introduction in the discussion of machine-learning-based diabetes prediction.

17. The Citation for Reference (Esteva et al. 2019) is missing in the manuscript.

Response: The in-text citation for Esteva et al. (2019) has been added in the Introduction in the broader context of machine learning/deep learning in healthcare.

We appreciate the opportunity to revise the submission and hope that the manuscript now satisfies the technical requirements.

Sincerely,
On behalf of the authors